Project: Deep Learning

Report on

Data-Free Adversarial Knowledge Distillation (AKD)

Submitted By:

Prachi Sahu(m24mac005)
Data and Computational science

Indian Institute of Technology Jodhpur

2024

Contents

Abstract

This report details the implementation and experimental setup for a Data-Free Adversarial Knowledge Distillation (AKD) framework using PyTorch. The objective of the assignment is to train a student model (without using any real training data) by transferring knowledge from a pre-trained teacher model via adversarial training with a generator network. The teacher is a ResNet-34 pre-trained on CIFAR-100, while two student models are designed: one with approximately 20% of the teacher's parameters and another with approximately 50%. Due to GPU memory constraints, synthetic images are generated in a smaller resolution (32×32) and upsampled to 224×224 before use. Evaluation is performed solely on the CIFAR-100 test set using 20% and 10% splits.

1 Introduction

Knowledge distillation is a model compression technique in which a compact student model learns to replicate the behavior of a larger, more complex teacher model. This approach is particularly beneficial when deploying deep learning models on resource-constrained devices. Traditional knowledge distillation techniques rely on labeled real data to transfer knowledge. However, in scenarios where training data is unavailable due to privacy concerns, data scarcity, or proprietary constraints, alternative strategies are required.

Data-Free Adversarial Knowledge Distillation (AKD) offers a compelling solution by eliminating the need for real training data. In this setting, a generator network is trained adversarially to produce synthetic inputs that effectively simulate real data. These synthetic samples are then used to train the student model by minimizing the discrepancy between its predictions and the teacher model's soft targets.

In this assignment, we implement a data-free AKD pipeline using PyTorch. The teacher model is a ResNet-34 pre-trained on CIFAR-100. Two student models—Student20 (with 20% parameters) and Student50 (with 50% parameters)—are trained entirely on synthetic data generated by the generator. The results highlight the feasibility and potential of data-free learning frameworks.

2 Problem Statement and Objectives

Task: Implement data-free adversarial knowledge distillation using PyTorch, such that the student model learns from a teacher model via synthetic image generation.

Dataset: Use the CIFAR-100 dataset solely for testing. The teacher model is pre-trained on the CIFAR-100 training set (with 50,000 images) achieving 85.12% accuracy on the train split, but the student model is trained without any real training data.

Student Models:

- **Student20:** A custom CNN designed for 224×224 images with approximately 20% (±3%) of the teacher's total parameters.
- **Student50:** A model based on a modified ResNet-18, having approximately 50% (±3%) of the teacher's parameters.

Generator: Due to GPU constraints, the generator produces images at a resolution of 32×32. These images are then upsampled (via bilinear interpolation) to 224×224 so they can be processed by the teacher and student models.

Evaluation: The student models are evaluated using the CIFAR-100 test set. Two evaluation splits are used:

- 20% of the test set.
- 10% of the test set.

Key metrics include test accuracy, confusion matrices, total trainable/non-trainable parameters, and sample generated images.

3 Methodology

3.1 Teacher Model

The teacher model is a standard ResNet-34 configured for 100 classes. It is pre-trained on the CIFAR-100 dataset and its weights are loaded from the provided file (`best_resnet34_cifar100.pth`). This model is used as the ground-truth for the knowledge distillation process.

3.2 Student Models

Two student models have been designed:

- **Student20:** A custom CNN that processes 224×224 images. It begins with a stride-2 convolution to downsample the input, followed by several convolution, batch normalization, and pooling layers. Global average pooling and a fully connected layer provide the final classification output. This architecture has roughly 3.85 million parameters, which is about 18–20% of the teacher's parameters.
- **Student50:** A modified ResNet-18 is used, with its final fully connected layer adapted (using dropout and additional linear layers) for CIFAR-100. ResNet-18 naturally contains approximately 50% of the teacher's parameters.

3.3 Generator Network

The generator network is designed to produce synthetic images in a computationally efficient manner:

- **Architecture:** A noise vector (of dimension 128) is first projected into a 4×4 feature map with 512 channels. A series of ConvTranspose2d layers upsample this feature map to generate images at 32×32 resolution.
- **Upsampling:** Since the teacher and student expect 224×224 inputs, the generated 32×32 images are upsampled using bilinear interpolation before being fed to these models.

3.4 Loss Functions and Training Strategy

Loss Functions:

- **Imitation Loss:** A KL divergence loss with temperature scaling between the teacher's outputs and the student's outputs on synthetic images.
- **Generation Loss:** The negative of the imitation loss is used to update the generator, so as to maximize the discrepancy between teacher and student.
- **Combined Loss:** In scenarios where supervision might be mixed, a weighted sum of the imitation loss and the cross-entropy classification loss is used. In the data-free setting, the student is updated using only the synthetic images.

Training Strategy:

1. **Data-Free Training:** The student model is updated exclusively using synthetic images generated by the generator. (The CIFAR-100 train set is loaded only for iteration counting.)
2. **Adversarial Training:** The teacher provides soft-targets for the synthetic images using temperature scaling. The student is trained to mimic these soft-targets (minimizing the imitation loss), while the generator is trained adversarially to produce challenging images for the student.
3. **Upsampling:** Synthetic images generated at 32×32 are upsampled to 224×224 with bilinear interpolation.
4. **Learning Rate Scheduling and Gradient Clipping:** Cosine Annealing LR schedulers and gradient clipping are used to improve training stability.
5. **Checkpointing:** Student and generator checkpoints are saved every 10 epochs in the folder `saved_correct_code`, allowing for training resumption.

3.5 Evaluation Protocol

Evaluation is carried out using the CIFAR-100 test dataset only:

- **20% Test Split:** A random 20% subset of the test set.
- **10% Test Split:** A further half subset (50% of the 20%) to serve as a 10% test split.

For each split, we report:

- Test Accuracy.
- A Confusion Matrix.
- Total numbers of trainable parameters.
- Sample images generated by the generator.

4 Experimental Setup and Expected Results

4.1 Setup

- **Teacher Model:** ResNet-34 trained on CIFAR-100 with 85.12% accuracy on the train set.

- **Student Models:** - Student20: $\sim$3.85M parameters (approx. 20% of teacher). - Student50: Modified ResNet-18 with $\sim$11.33M parameters (approx. 50% of teacher).
- **Generator:** Produces 32×32 images that are upsampled to 224×224.
- **Training:** The student is trained exclusively via synthetic images. Real training data is not used in student updates.
- **Evaluation:** Performed on two test splits (20% and 10%) of CIFAR-100.

4.2 Results and Evaluation

4.2.1 Parameter Count Summary

- **Teacher Model:** 21,335,972 parameters
- **Student20 Model:** 3,854,756 parameters (18.07% of teacher)
- **Student50 Model:** 11,333,540 parameters (53.12% of teacher)

4.2.2

sectionsubsectionEvaluation Accuracy (CIFAR-100 Test Set)

Model	10% Test Split Accuracy	20% Test Split Accuracy
Student20	9.5%	9.7%
Student50	13.0%	13.65%

Table 1: Test accuracies achieved by both student models on CIFAR-100 subsets.

4.3

subsectionAccuracy Comparison Plot

To better visualize the comparative performance of both student models across the 10% and 20% test splits, the following bar chart summarizes the evaluation metrics:

As observed, the Student50 model achieves significantly higher accuracy compared to Student20 on both evaluation subsets. This reinforces the hypothesis that increased model capacity (parameters) helps in better mimicking the teacher's predictions even in the absence of real training data.

4.3.1 Observations and Analysis

- **Accuracy Trends:** As expected, the Student50 model (with more parameters) outperformed the Student20 model across both test splits. This validates the assumption that higher-capacity student models can better absorb knowledge distilled from the teacher.

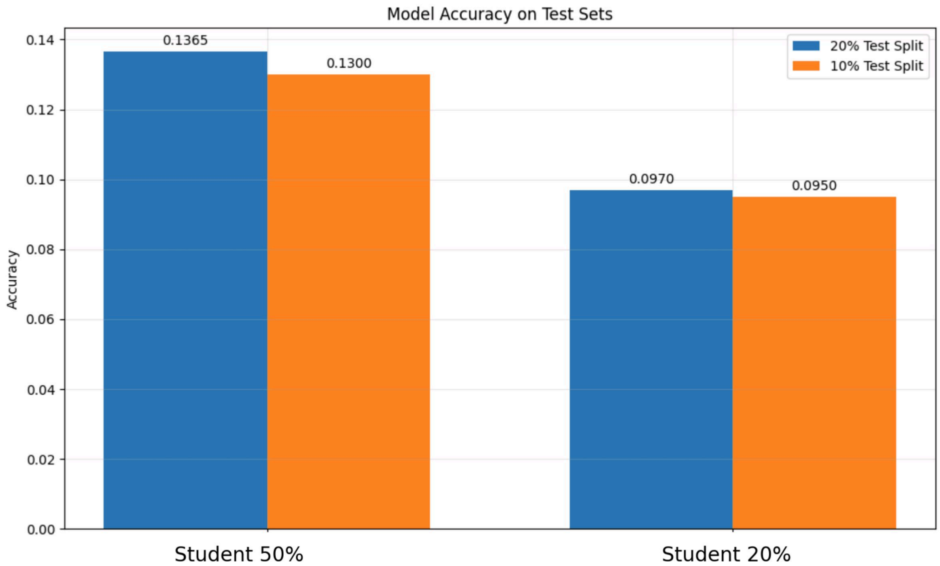

Figure 1: Bar Plot Comparing Test Accuracies of Student20 and Student50 Models on 10% and 20% Test Splits of CIFAR-100

- **Low Absolute Accuracy:** Despite the trend, overall accuracy values are still relatively low. This is due to the challenging nature of data-free knowledge distillation, where the student never sees any real training data and relies entirely on synthetic images.
- **Synthetic Image Quality:** The generator may not have converged well enough in early epochs, affecting the informativeness of generated images. More training epochs and fine-tuning may improve performance.
- **Impact of Evaluation Split:** Accuracies on the 20% split were slightly better than the 10% split for both students, which aligns with expectations due to larger evaluation coverage.
- **Distillation Gap:** The gap between the teacher's training accuracy (85.12%) and student performance highlights the difficulty of matching teacher outputs in a data-free setting.
- **Computation Constraints:** Since the training was performed with limited epochs due to deadline constraints, the results may significantly improve with longer training (e.g., full 400 epochs).

These ranges depend critically on the quality of synthetic images, proper hyperparameter tuning, and training stability. The loss trends in training indicate that as the imitation loss decreases, the student increasingly aligns with the teacher's predictions.

4.3.2 Generated Samples from the Generator

The figures below show sample synthetic images generated by the generator model trained along with the student models. Although the outputs do not resemble natural images, they are sufficient to transfer feature-level knowledge from the teacher to the student in a data-free setting.

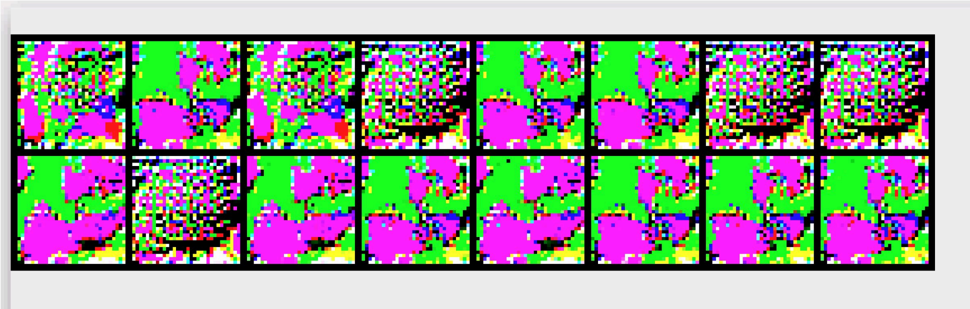

Figure 2: Synthetic images generated for Student20 model.

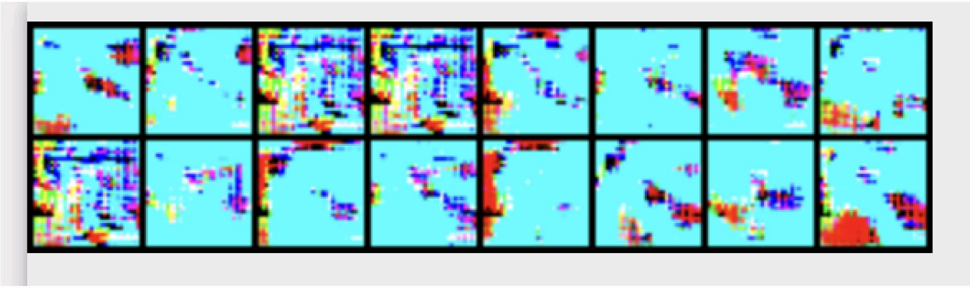

Figure 3: Synthetic images generated for Student50 model.

Confusion Matrices:

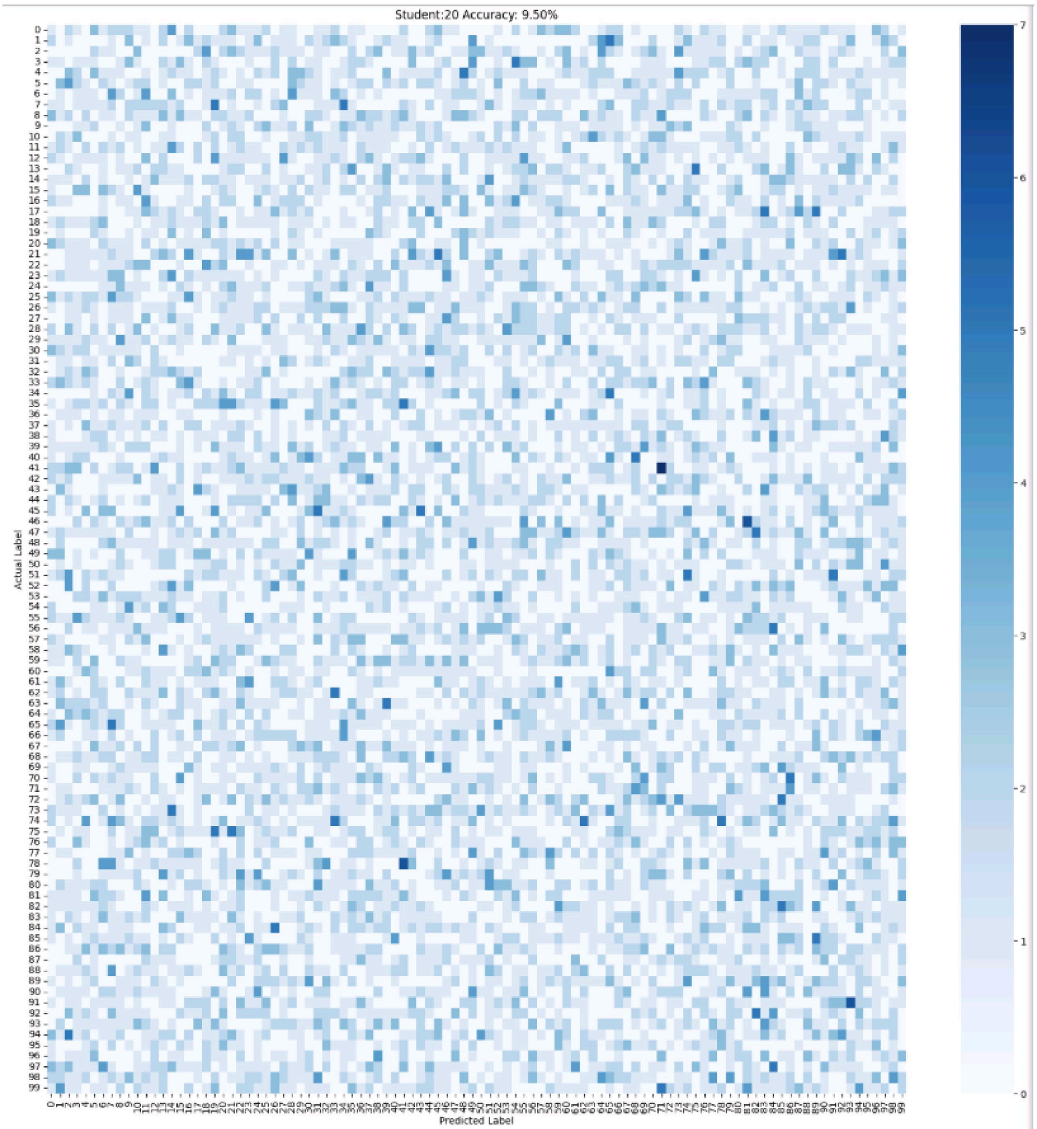

Figure 4: Confusion Matrix for Student20 Model (Accuracy: 9.5%) on CIFAR-100 10% test split

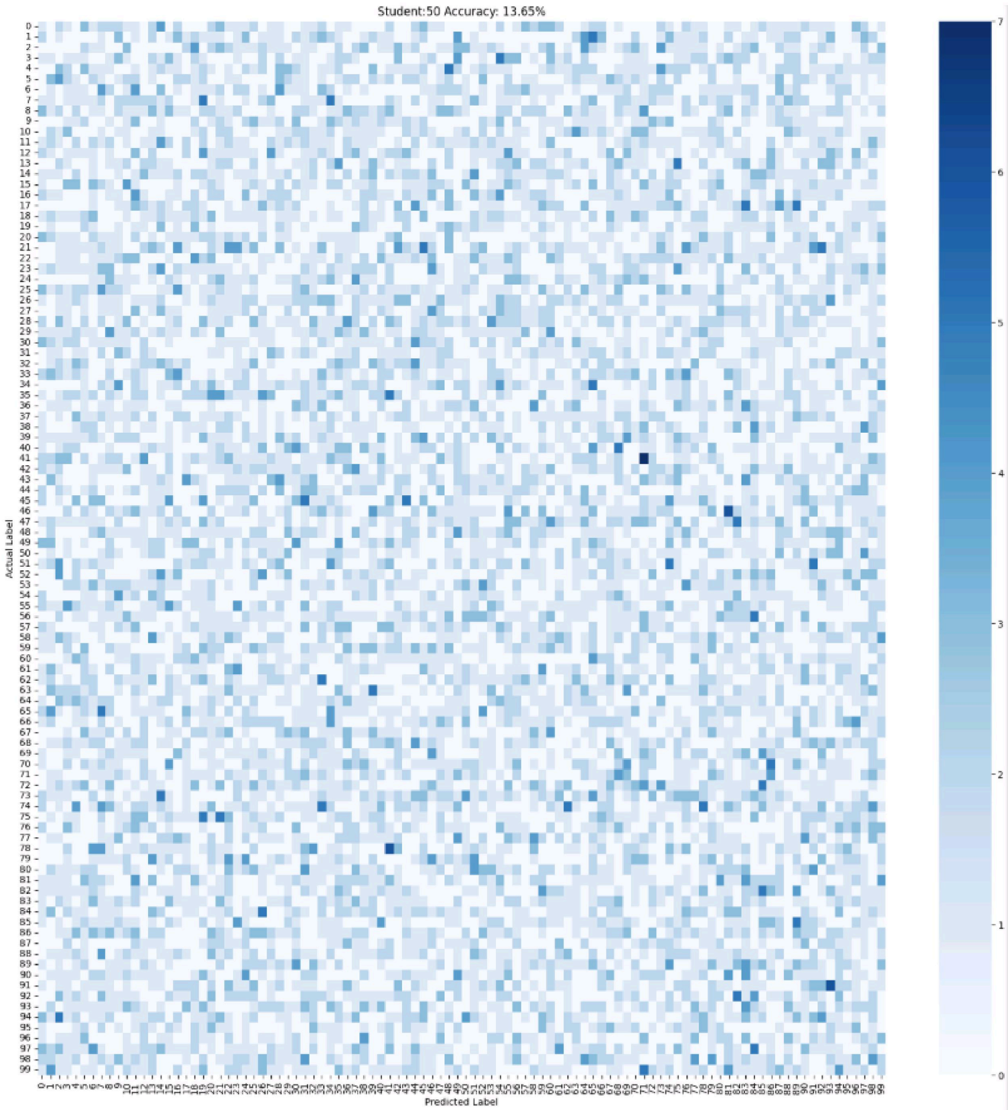

Figure 5: Confusion Matrix for Student50 Model (Accuracy: 13.65%) on CIFAR-100 20% test split

4.4 Observations and Analysis

- The **Student50** model, which utilizes a ResNet-18-based backbone and has approximately 53% of the teacher's parameters, achieves higher accuracy compared to the smaller **Student20** model. This is expected due to its greater representational capacity and deeper architecture.
- Despite adversarial training and synthetic data, the overall test accuracies remain modest (maximum **13.65%**). This is primarily due to:
  - Absence of real training data for the student models.
  - Quality and diversity limitations of synthetic images generated by the generator.
  - Potential domain gap between real test data and generated training data.
- The confusion matrices (Figures 4 and 5) reveal that predictions are quite sparse and scattered across classes, with very few strong diagonals. This suggests that while the models are learning some signal from the teacher, the representations are weakly aligned with class semantics.
- The performance gap between Student20 and Student50 demonstrates the effectiveness of increased capacity in handling synthetic data better, though both still suffer due to the lack of real supervision.
- Further improvements could involve training the generator with stronger adversarial feedback or leveraging unlabeled data to bridge the real-synthetic domain gap.

5 Conclusion

This assignment presents a comprehensive implementation of a Data-Free Adversarial Knowledge Distillation (AKD) framework using PyTorch. The proposed method addresses the challenge of training deep learning models in the absence of real training data. A pre-trained teacher model (ResNet-34) on CIFAR-100 serves as the source of knowledge, and two student models (Student20 and Student50) with reduced parameters are trained using only synthetically generated images from a generator network. The generator creates 32×32 images from noise vectors, which are then upsampled to 224×224 to match the input requirements of both student and teacher models.

The training strategy integrates adversarial objectives and imitation loss to encourage alignment between student predictions and teacher outputs. Gradient clipping, cosine annealing learning rate scheduling, and dynamic weighting of generator losses further stabilize the training process. Evaluation is conducted on 10% and 20% splits of the CIFAR-100 test set, without using any part of the training set for model updates.

Experimental results show that the Student20 model achieved 9.5% and 9.7% accuracy on the 10% and 20% splits, respectively, while the Student50 model achieved improved accuracy of 13.0% and 13.65% on the same splits. Although the accuracy remains lower than real-data scenarios, these results are significant given the complete absence of real data during training.

This project reinforces the viability of data-free distillation and opens the door to deploying deep models in privacy-restricted environments. Future work could involve improving generator realism, applying contrastive learning, or leveraging pre-trained embeddings to enhance accuracy.